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Integrating Financial Management and Gamification: A Systematic Literature Review and Future Research Agenda

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Abstract

The implementation of gamification in personal financial management (PFM) apps has the potential to motivate users to engage in positive financial behaviors. However, several research gaps corroborate the need to investigate gamification in the financial domain further. To address these gaps, this systematic literature review comprehensively examines current research on financial behavior and gamification. This review utilizes the Theory, Context, Characteristics, and Methodology (TCCM) framework to provide a holistic understanding of financial behavior and gamification-related behavioral intention. A total of 53 articles published between 2018 and 2022 were analyzed to assess the present research landscape and provide directions for future studies. This review makes three key contributions: (1) synthesizes current research on financial behavior and gamification-related behavioral intention, (2) presents integrated conceptual models that elucidate financial behavior and gamification-related behavioral intention, and (3) identifies research gaps and suggests avenues for future investigation.

A. Introduction

In recent years, there has been growing concern regarding the low level of public financial behavior [1]. Researchers have found that many individuals make bad financial decisions, save too little, spend too much [2], and invest in things they later regret [3]. Financial behavior has a significant impact on an individual's long-term financial outcomes and can lead to long-term financial problems [1], [4]–[6].

The financial sector is among the first to adopt mobile technology [7], [8]. Along with the rise of mobile banking and payment apps [9], personal financial management (PFM) apps have emerged as one of the fastest-growing categories of financial applications [7]. In addition to the rise in mobile banking and payment apps [9], PFM apps are one of the fastest-growing categories of financial applications [7]. The global PFM application market is expected to reach $3,338 million by 2025 with an annual growth rate of 12.65 % from 2018 [10].

However, there is a current challenge in effectively integrating technology to enhance the quality of life [11]–[13]. One way to improve user experience, motivation, and interest is through gamification, the implementation of game features and game-like designs in apps [7], [14]. Gamification has gained significant attention from academic researchers and practitioners [15], [16]. However, most of the existing research related to gamification has focused on the education and learning sector, while other sectors, such as finance, have received limited attention [7], [17]–[19]. This highlights a research gap in gamification in personal finance, despite its recognized importance [15] and widespread adoption [7], [20].

Previous reviews have only examined financial behavior [1] or gamification [21]. Therefore, it is beneficial to address these research gaps by conducting a study that reviews and synthesizes the relationship between these two topics. This study employs a systematic literature review methodology to map the current financial behavior and gamification research state. Specifically, this study aims to bridge the gap between financial behavior with gamification by reviewing research on gamification-related behavioral intention.

This review employs the Theory, Context, Characteristics, and Methodology (TCCM) framework developed by [22]. The TCCM framework provides more comprehensive and robust insights compared to other approaches [23], [24], enabling a holistic understanding of a research domain [25]. Furthermore, the TCCM framework allows for theoretical and empirical mapping of a research field [25]–[27]. This analysis includes 53 articles to review the current state of research and provide directions for future research. Overall, this review provides three contributions by (1) providing a holistic synthesis of existing research on financial behavior and gamification-related behavioral intention, (2) presenting integrated conceptual models that explain financial behavior and gamification-related behavioral intention, and (3) identifying research gaps and providing directions for future research.

The remainder of the paper is structured as follows. Section 2 explains the methodology employed in this systematic literature review. Section 3 presents this review's results and future research agendas based on the TCCM framework. Finally, Section 4 concludes the study.

B. Research Method

The systematic literature review was conducted following the methodological guidelines outlined by [28]. This study aims to identify, evaluate, and interpret existing research in the domains of financial behavior and gamification-related behavioral intention using the TCCM framework [25]. The search strategy, study selection process, and data extraction methods are deescribed as follows.

Search Strategy

Four reputable online research databases were selected: Emerald Insights, IEEE Xplore, Science Direct, and Taylor & Francis. For each domain, specific search strings were developed incorporating relevant key terms. In addition to core terms related to financial behavior and gamification, terms such as "theory," "framework," and "model" were included. The resulting search strings were as follows.

("personal") AND ("financial" OR "finance") AND ("management" OR "managing") AND ("behavior" OR "behavior") AND ("theory" OR "theoretical" OR "framework" OR "model")

("gamification" OR "gamify" OR "gamified") AND ("theory" OR "theoretical" OR "framework" OR "model")

The search was conducted in September 2022, limited to article titles, abstracts, and keywords. Only journal articles and proceedings papers published between 2018 and 2022 were included. This search resulted in 27,888 articles: 20,078 articles on financial behavior and 7,810 articles on gamification-related behavioral intention.

Study Selection

The study selection process consisted of title and abstract selection, followed by full-text selection. Throughout both steps, articles were assessed based on the following inclusion criteria: (1) articles written in English, (2) articles that investigated either financial behavior or gamification-related behavioral intention, (3) empirical research presenting a research model or hypotheses, and (4) articles that were not literature reviews. Fifty six relevant articles were identified in the first step: 27 on financial behavior and 29 on gamification-related behavioral intention. In the second step, three articles on financial behavior were excluded, resulting in a final selection of 53 articles.

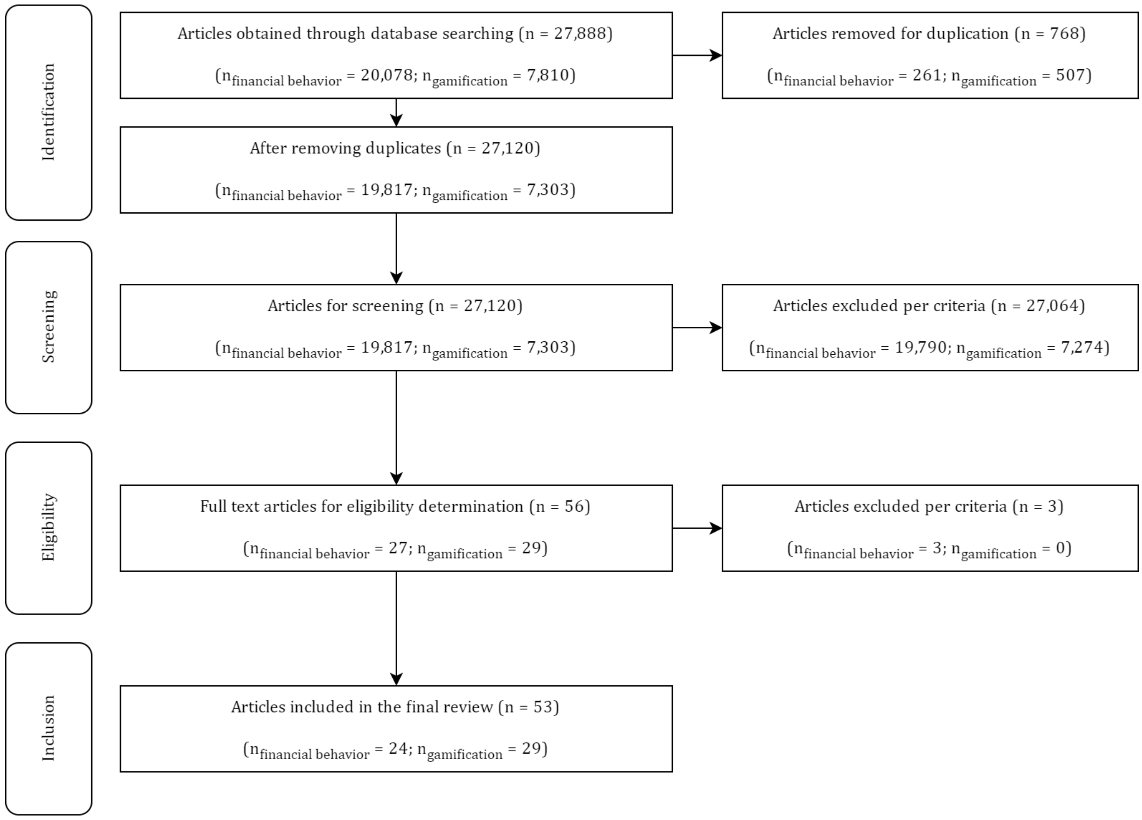

Figure 1. Study selection process

Data Extraction

This systematic literature review followed the domain-based approach using the TCCM framework developed by [22]. Relevant publication details such as title, year of publication, and publication outlet were extracted. Furthermore, information related to theories, contexts, characteristics or constructs, and research methodologies was extracted following the TCCM framework. Finally, research limitations and suggestions provided by the authors for future research were extracted to identify potential research directions.

C. Result and Discussion

Overview of Reviewed Articles

This section organizes the articles based on their publication year and outlet. Figure 2 illustrates the growth of research on financial behavior and gamification-related behavioral intention from 2018 to 2022. It is evident that research interest in both domains has been increasing in recent years.

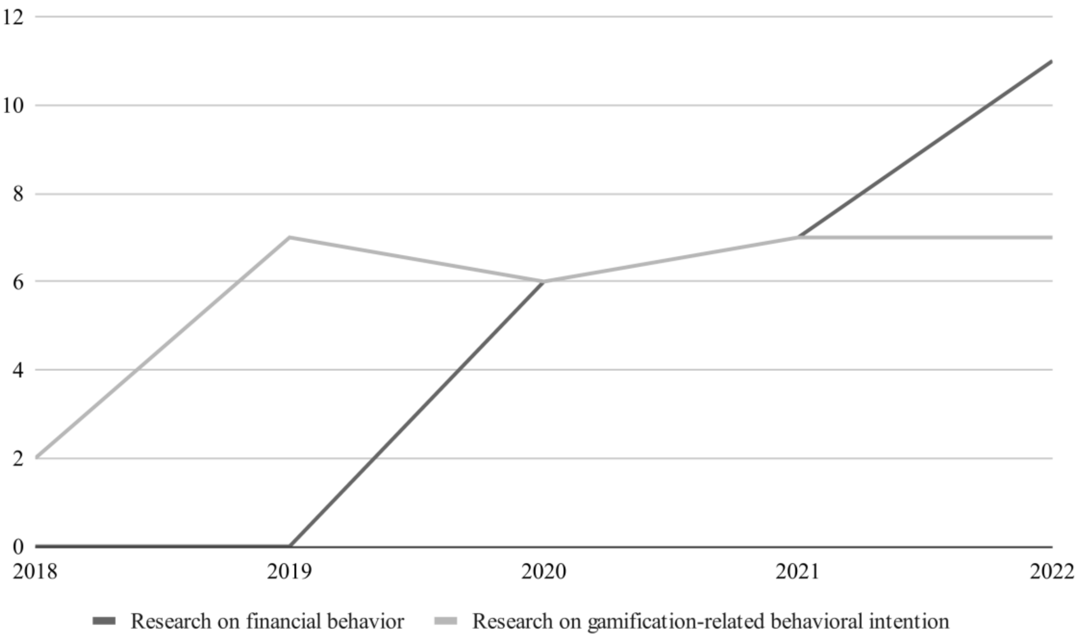

Figure 2. Number of articles by year of publication

Furthermore, Table 1 presents the finance, business, and management journals as publication outlets for articles on financial behavior. On the other hand, Table 2 shows that articles on gamification-related behavioral intention have been published in various journals, including those focused on human-computer interaction, information systems, management, computer science, education, and business. Some articles on gamification-related behavioral intention were also sourced from management and education technology proceedings.

Table 1. Publication Outlets of Financial Behavior Research

Publication Outlet	SJR	#	%	References
International Journal of Bank Marketing	Q2	4	16.7	[29]–[32]
Young Consumers	Q1	3	12.5	[15], [33], [34]
Cogent Business and Management	Q2	3	12.5	[4], [35], [36]
Managerial Finance	Q3	2	8.3	[37], [38]
Applied Developmental Science	Q1	1	4.2	[39]
Economics of Innovation and New Technology	Q1	1	4.2	[40]
International Journal of Sociology and Social Policy	Q1	1	4.2	[41]
Journal of Services Marketing	Q1	1	4.2	[42]
Asia-Pacific Journal of Business Administration	Q2	1	4.2	[43]
Cogent Social Sciences	Q2	1	4.2	[44]
Journal of Behavioral and Experimental Economics	Q2	1	4.2	[45]
Kybernetes	Q2	1	4.2	[46]
Marriage and Family Review	Q2	1	4.2	[47]
Review of Behavioral Finance	Q2	1	4.2	[48]
Journal of Workplace Behavioral Health	Q3	1	4.2	[49]
Journal of Money and Business	-	1	4.2	[50]

Table 2. Publication Outlets of Gamification-Related Behavioral Intention Research

Publication Outlet	SJR	#	%	References
International Journal of Human-Computer Interaction	Q1	3	10.3	[51]–[53]
Industrial Management and Data Systems	Q1	2	6.9	[54], [55]
Information Technology and People	Q1	2	6.9	[11], [56]
International Journal of Human Computer Studies	Q1	2	6.9	[57], [58]
Computers in Human Behavior	Q1	1	3.4	[59]
Enterprise Information Systems	Q1	1	3.4	[60]
International Journal of Information Management	Q1	1	3.4	[61]
Internet Research	Q1	1	3.4	[62]
Journal of Business Research	Q1	1	3.4	[63]
Journal of Enterprise Information Management	Q1	1	3.4	[64]
Journal of Further and Higher Education	Q1	1	3.4	[65]
Journal of Service Theory and Practice	Q1	1	3.4	[66]
Journal of Small Business and Enterprise Development	Q1	1	3.4	[67]
Young Consumers	Q1	1	3.4	[68]
Education and Training	Q2	1	3.4	[69]
Interactive Technology and Smart Education	Q2	1	3.4	[70]
International Journal of Bank Marketing	Q2	1	3.4	[7]
International Journal of Sports Marketing and Sponsorship	Q2	1	3.4	[71]
Journal of Systems and Information Technology	Q2	1	3.4	[72]
Learning and Motivation	Q2	1	3.4	[73]
World Journal of Science, Technology and Sustainable Development	-	1	3.4	[74]
2019 16th International Conference on Service Systems and Service Management, ICSSSM 2019	-	1	3.4	[75]
2020 8th International Conference on Cyber and IT Service Management, CITSM 2020	-	1	3.4	[76]
Proceedings - IEEE 10th International Conference on Technology for Education, T4E 2019	-	1	3.4	[77]

Theory

Upon reviewing the literature, it was found that six theories have been employed as theoretical foundations to investigate financial behavior. The theory of planned behavior emerged as the most frequently used theory, accounting for 50% of the studies. Other theories utilized included the family stress theory (16.7%), behavioral life-cycle theory (8.3%), family financial socialization theory (8.3%), management system theory (8.3%), and self-determination theory (8.3%).

While the reviewed studies on financial behavior did not provide specific suggestions for future theory development, a systematic literature review on personal financial management behavior [1] offers potential avenues for theoretical exploration. This review suggests employing interdisciplinary theoretical approaches to examine how individuals' financial behavior is influenced by different life stages [78], [79]. [1] also proposes investigating the connections between financial education, personal financial management behavior, and life satisfaction [79]–[81]. Additionally, socio-psychological theories could be developed to understand behavioral biases from cultural and social factors [1]. Finally, [1] also recommends employing the theory of planned behavior and the transtheoretical model in investigating diverse behaviors and populations [29].

Table 3. Theories Used in Financial Behavior Research

Theory	#	%	References
Theory of Planned Behavior	6	50	[4], [29], [32], [36], [43], [45]
Family Stress Theory	2	16.7	[47], [49]
The Behavioral Life-Cycle Theory	1	8.3	[50]
Family Financial Socialization Theory	1	8.3	[31]
Management System Theory	1	8.3	[47]
Self-Determination Theory	1	8.3	[33]

On the other hand, 25 theories were identified in studies exploring gamification-related behavioral intention. The majority of these theories (54%) focused on technology adoption. Notably, the self-determination theory (14.3%), technology acceptance model (11.9%), theory of planned behavior (9.5%), expectation-confirmation model (7.1%), and the unified theory of acceptance and use of technology (7.1% for both version 1 and 2) were the most frequently employed. Other theories related to technology use include the continuance theory, diffusion of innovation, flow theory, hedonic motivation system adoption model, uses and gratification theory, and user engagement theory. In addition, there were gamification-specific theories such as the gamification theory, Landers' theory of gamified learning, and work gamification theory. The remaining theories used in gamification research pertained to psychology (e.g., cognitive absorption theory, cognitive evaluation theory, goal framing theory, goal priming theory, theory of reasoned action, prospect theory, theory of perceived risk, and theory of consumption values) and social theories (e.g., generational cohort theory and social identity theory).

Regarding theoretical development opportunities in gamification-related behavioral intention research, future studies can employ relevant theories to investigate adoption, reuse, and continuance intention [56], [59]. Motivation theory is also suggested as a viable option [53], [82]. Additionally, using existing or new theories on dynamic and cyclical gamification is recommended [21]. Exploring potential similarities between constructs from different theories over time is another avenue for investigation [53].

Table 4. Theories Used in Gamification-Related Behavioral Intention Research

Theory	#	%	References
Self-Determination Theory	6	14.3	[7], [51], [57], [59], [67], [70]
Technology Acceptance Model	5	11.9	[7], [11], [52], [70], [73]
Theory of Planned Behavior	4	9.5	[52], [67], [73], [77]
Expectation-Confirmation Model	3	7.1	[56], [64], [69], [75]
Unified Theory of Acceptance and Use of Technology 2	3	7.1	[53], [71], [72]
Unified Theory of Acceptance and Use of Technology	2	4.8	[68], [74]
Cognitive Absorption Theory	1	2.4	[64]
Cognitive Evaluation Theory	1	2.4	[63]
Continuance Theory	1	2.4	[51]
Diffusion of Innovation	1	2.4	[74]
Flow Theory	1	2.4	[69]
Generational Cohort Theory	1	2.4	[72]
Gamification Theory	1	2.4	[72]

Goal Framing Theory	1	2.4	[54]
Goal Priming Theory	1	2.4	[60]
Hedonic Motivation System Adoption Model	1	2.4	[76]
Landers' Theory of Gamified Learning	1	2.4	[65]
Prospect Theory	1	2.4	[11]
Social Identity Theory	1	2.4	[51]
Theory of Consumption Values	1	2.4	[62]
Theory of Reasoned Action	1	2.4	[73]
Theory of Perceived Risk	1	2.4	[62]
User Engagement Theory	1	2.4	[55]
Uses and Gratification Theory	1	2.4	[68]
Work Gamification Theory	1	2.4	[55]

While it is evident that a wide range of theories have been utilized to analyze gamification, there is a narrower selection of theories applied in financial behavior research. Nevertheless, the theory of planned behavior, commonly employed in financial behavior studies, is also relevant to gamification research. Furthermore, the self-determination theory demonstrates relevance in both domains and has the potential for increased utilization in financial behavior research.

Context

Table 5 summarizes the financial contexts covered in the reviewed articles investigating financial behavior. Most research in this area generally focuses on financial behavior (21%). Other aspects of financial behavior that have been studied include responsible financial behavior, saving behavior, spending behavior, and retirement behavior. Additionally, common research areas include financial literacy (10.5%), financial satisfaction (7.9%), financial well-being (7.9%), financial goals (5.3%), and financial stress (5.3%).

Future research on financial behavior is encouraged to explore previously studied financial behaviors in different countries, cultures, and demographics [45]. Furthermore, investigating financial behavior across various life stages [31], including the transitional period of young adults and their partners [39], [83], is recommended. Additionally, research can focus on financial behavior within different financial sectors, such as banking, insurance, and securities markets [38].

Table 5. Contexts of Financial Behavior Research

Context	#	%	References
General financial behavior	8	21	[4], [15], [30], [33], [39], [44]–[46]
Financial literacy	4	10.5	[37], [38], [40], [43]
Financial satisfaction	3	7.9	[36], [39], [47]
Financial well-being	3	7.9	[36], [42], [48]
Financial goals	2	5.3	[35], [39]
Financial stress	2	5.3	[4], [49]
Responsible financial behavior	2	5.3	[29], [34]
Saving behavior	2	5.3	[40], [41]
Attitudes	1	2.6	[45]
Depositor discipline	1	2.6	[37]
Family financial socialization	1	2.6	[31]
Financial confidence	1	2.6	[44]
Financial knowledge	1	2.6	[44]
Investment decision	1	2.6	[50]
Learning capacity	1	2.6	[44]

Retirement behavior	1	2.6	[35]
Retirement planning	1	2.6	[32]
Self-control	1	2.6	[45]
Spending behavior	1	2.6	[40]
Workplace productivity	1	2.6	[49]

Meanwhile, Table 6 presents the contexts of the reviewed articles investigating gamification-related behavioral intention. These articles explore gamification in education (20.7%), finance (17.2%), games and gamified systems in general (17.2%), health and fitness (17.2%), e-commerce (6.9%), and management (6.9%). Furthermore, gamification research has also started to extend into business, entertainment, environmental, and marketing contexts. Table 7 summarizes the gamification elements investigated or implemented in these reviewed articles. The analysis reveals the involvement of various gamification elements, encompassing achievement-related, social-related, and immersion-related aspects.

Future research on gamification-related behavioral intention is suggested to explore new sectors [72] or types of gamified apps [56]. Specifically, investigating specific gamified apps rather than considering an entire app category as a whole is encouraged [56]. Analyzing these new contexts can deepen the understanding of gamification implementation methods and outcomes [63]. Additionally, future research can explore new gamification elements and the consequences of each element [57], [72]. Elements of gamification that can be further explored include narrative, mechanics, aesthetics, and technological aspects [68]. As mentioned in the systematic literature review by [21], future research can also investigate cooperative and collective gamification.

Table 6. Contexts of Gamification-Related Behavioral Intention Research

Context	#	%	References
Education	6	20.7	[51], [57], [65], [69], [70], [77]
Finance	5	17.2	[7], [11], [64], [72], [74]
Games and general gamified systems	5	17.2	[53], [56], [58], [59], [73]
Health and fitness	5	17.2	[52], [60]–[62], [71]
E-commerce	2	6.9	[68], [75]
Management	2	6.9	[55], [59]
Business	1	3.4	[67]
Entertainment	1	3.4	[76]
Environmental	1	3.4	[54]
Marketing	1	3.4	[66]

Table 7. Gamification Elements Investigated in Gamification-Related Behavioral Intention Research

Gamification Element	#	%	References
Badge	6	9.7	[55], [57], [67], [71], [73], [77]
Point	6	9.7	[55], [64], [67], [69], [72]
Games	5	8.0	[53], [64], [68]–[70]
Leaderboards	4	6.4	[53], [55], [67], [73]
Monitoring	4	6.4	[52], [61], [66], [71]
Social	4	6.4	[52], [61], [71], [75]
Challenge	3	4.8	[66], [67], [77]

Feedback	3	4.8	[59], [66], [77]
Levels	3	4.8	[55], [61], [67]
Training	3	4.8	[66], [67], [71]
Achievements	2	3.2	[53], [54]
Competition	2	3.2	[54], [73]
Goal	2	3.2	[52], [73]
Prizes	2	3.2	[72], [73]
Rewards	2	3.2	[52], [72]
Autonomy support	1	1.6	[54]
Avatar	1	1.6	[59]
Character	1	1.6	[66]
Cooperation	1	1.6	[73]
Gifting	1	1.6	[11]
Immersion	1	1.6	[75]
Interactivity	1	1.6	[54]
Medals	1	1.6	[61]
Narrative	1	1.6	[60]
Storytelling	1	1.6	[73]
Time constraint	1	1.6	[73]

The results also indicate that several studies examine gamification in financial contexts. These studies are conducted by [7], [11], [64], [72], [74]. Most of these studies focus on gamification in banking contexts, as seen in the works of [74], [72], and [11]. Additionally, a study by [64] specifically explores gamification in robo-advisor apps, while a study by [7] concentrates on gamification in the PFM app Mint.

Characteristics

This review encompasses a range of factors that influence financial behavior and gamification-related behavioral intention, including antecedents, mediators, moderators, and consequences. Figures 3 and 4 illustrate integrated conceptual models depicting financial behavior and gamification-related behavioral intention. We further synthesize these constructs in detail.

Antecedents

Financial-related constructs
- Financial literacy
- Financial attitude
- Financial knowledge
- Financial attainment capacity
- Financial efficacy
- Financial planning activity
- Financial socialization
- Financial stress
- Financial self efficacy
- Financial skills
- Retirement benefit sufficiency

Psychological constructs
- Locus of control
- Optimism
- Autonomy
- Competence
- Deliberative thinking
- Neurotic personality
- Relatedness
- Future orientation
- Materialism as centrality
- Materialism as happiness
- Materialism as success

Socio-demographic constructs
- Education
- Gender
- Age
- Income
- Marital status
- Ethnicity
- Social-economic standing

External constructs
- Parents' teaching & behavior
- Perceived health status
- Perceived parental status
- Relevant hazard experience

Mediators

Psychological constructs
- Habit
- Learning capacity
- Perceived behavioral control
- Perceived financial information insufficiency
- Perceived hazard characteristics
- Risk aversion
- Affective impulsiveness
- Affective response
- Cognitive impulsiveness

Financial-related constructs
- Financial anxiety
- Financial confidence
- Financial description
- Financial inclusion
- Financial security
- Personal financial distress
- Retirement planning strategies

External constructs
- Informational subjective norms

Socio-demographic constructs
- Dependents
- Occupation

Moderators
- Financial risk tolerance
- Gamifying features
- PFM product

Behaviors
- Financial behavior
- Saving behavior
- Spending behavior
- Investment behavior
- Behavior
- Borrow
- Debt behavior
- Debt ratio change
- Financial information-seeking (FIS) behavior & avoidance
- Herding behavior
- Indebtedness behaviors
- Plan
- Responsible financial behavior (RFB)
- Retirement behavior

Moderators
- Financial satisfaction
- Short-term financial goals

Consequences
- Financial well-being
- Depressive symptoms
- Financial obstacles to goal attainment
- Future saving foresight
- Future spending foresight
- Happiness
- Investment decisions
- Long-term financial goals
- Market discipline
- Workplace productivity

Figure 3. Integrated conceptual model of financial behavior

Antecedents

Gamifcation-specific constructs
- Gamification
- Avatar similarity
- Badge characteristics
- Challenge
- Game attribute
- Gameful design processes
- Gamification narrative design
- Competition
- Coaching
- Quantified self
- Sharing
- Tracking
- Virtual training
- Visibility of achievement
- Embodied feedback
- Feedback

Social constructs
- External regulation
- Perceived network size
- Social
- Social interaction
- Social need
- Social networking
- Social value

Context-specific constructs
- Financial risk
- General health
- Learning
- Context characteristics
- Physical appearance

Socio-demographic constructs
- Age
- Education
- Gender
- Years of experience

Psychological constructs
- Enjoyment
- Hedonic motivation
- Affective need
- Affiliation
- Autonomy support
- Behavioral beliefs
- Behavior monitoring
- Cognitive need
- Condition
- Control beliefs
- Curiosity
- External regulation
- Hedonic engagement
- Identified regulation
- Immersion
- Instrumental engagement
- Introjected regulation
- Meaningful engagement
- Motivational affordances
- Normative beliefs

Mediators

Psychological constructs
- Competence
- Perceived behavioral control
- Autonomous motivation
- Autonomy
- Cognitive absorption
- Cognitive Evaluation Theory (CET) needs satisfaction
- Compatibility
- Control
- Curiosity
- Decision freedom
- Identification
- Intrinsic motivation
- Joy
- Knowledge
- Motivation
- Other psychological effects of the system
- Perceived concentration
- Perceived control
- Perceived curiosity
- Perceived understanding
- Real-world purpose
- Regret
- Relatedness
- Relatedness satisfaction

External constructs
- Social influence
- Subjective norms
- Facilitating conditions
- Organizational identity
- Social facilitation
- Social gain
- Social relatedness

Socio-demographic constructs
- Age
- Education
- Gender
- Years of experience

Gamification-specific constructs
- Autonomous motivation
- Autonomy
- Gameful system characteristics
- Goal setting
- Perceived challenge
- Perceived effectiveness of gamification
- Rewards
- Self-monitoring
- Task diagnosticity
- Task meaningfulness
- User gameful experience

Context-specific constructs
- Green effectiveness

Technological constructs
- Attitude
- Satisfaction
- Perceived ease of use
- Perceived usefulness
- Effort expectancy
- Performance expectancy
- Confirmation
- Apps characteristics
- Apps design
- Flow experience
- Perceived benefits

Moderators

Gamifcation-specific constructs
- Autonomous motivation
- Autonomy
- Gameful system characteristics
- Goal setting
- Perceived challenge
- Perceived effectiveness of gamification
- Rewards
- Self-monitoring
- Task diagnosticity
- Task meaningfulness
- User gameful experience
- Achievement-related gamification elements
- Affective feedback
- Immersion-related gamification elements
- Informational feedback
- Social-related gamification elements

Socio-demographic constructs
- Gender differences
- Generation

External constructs
- Situational context
- Social feedback

Psychological constructs
- User psychological knowledge, skills, abilities, and other characteristics (KSAOs)

Technological constructs
- Perceived risks

Behaviors
- Behavioral intention
- Continuance intention
- Intention to use
- Intention to recommend
- Discontinuance intention
- Intention
- Persistent usage intentions
- Reuse Intention

Consequences
- Behavior
- Learning outcome
- Continued use
- Distal system-level change
- Perceived impact on learning
- Use behavior
- User behavioral change

Figure 4. Integrated conceptual model of gamification-related behavioral intention

1) Antecedents — Antecedents of financial behavior can be categorized into several groups: financial-related, socio-demographic, psychological, and external constructs. Financial-related antecedents, such as financial literacy, attitude, and knowledge, have been widely employed to investigate financial behavior.

Furthermore, socio-demographic constructs, including education, gender, age, income, and marital status, have demonstrated significant relevance in prior research. Additionally, a psychological construct that bears relevance is self-control. Turning to antecedents of gamification-related behavioral intention, they can be classified into psychological, socio-demographic, social, gamification, and context-specific constructs. Notably, these constructs commonly revolve around hedonic and affective aspects. Overall, intersecting constructs in financial behavior and gamification research pertain to education, autonomy, control, and self-efficacy, thereby warranting further exploration in the investigation of financial behavior and gamification.

2) Mediators of antecedents and behavior —Mediators of financial behavior and its antecedents can be categorized into psychological, socio-demographic, financial-related, and external constructs. Among these, habit is the most commonly observed mediating construct between financial behavior and its antecedents. On the other hand, mediators of gamification-related behavioral intention and its antecedents can be classified into psychological, technological, socio-demographic, external gamification, and context-specific constructs. These mediators are predominantly derived from technology adoption theories such as the theory of planned behavior, technology acceptance model, unified theory of acceptance and use of technology, expectation-confirmation model, and self-determination theory. They encompass factors such as attitude, satisfaction, perceived ease of use or effort expectancy, perceived usefulness or performance expectancy, confirmation, social influence, competence, perceived behavioral control, subjective norms, and facilitating conditions.

3) Moderators of antecedents and behavior — Moderators of financial behavior and its antecedents encompass variables such as financial risk tolerance, gamifying features, and PFM products. Meanwhile, moderators of gamification-related behavioral intention and its antecedents primarily revolve around constructs associated with gamification. Besides gender differences, moderators include achievement-related gamification elements, affective feedback, immersion-related gamification elements, informational feedback, social feedback, and social-related gamification elements. Considering these findings, it is evident that gamifying features and PFM products influence financial behavior. Given this connection between financial behavior and gamification, further exploration of both constructs is warranted and holds a substantial interest.

4) Behavior — This review encompasses a range of constructs that capture both financial behavior and gamification-related behavioral intention. In the realm of financial behavior, various constructs have been examined, including saving, spending, investment, debt, herding, and retirement behavior. On the other hand, in the context of gamification-related behavioral intention, in addition to behavioral intention itself, other variables that have been investigated include continuance intention, intention to use, discontinuance intention, persistent usage intentions, reuse intention, and intention. Notably, when comparing the research on financial behavior and gamification, it becomes evident that the current focus of gamification research leans more toward behavioral intention rather than the actual observed behavior.

5) Mediators of behavior and consequences — This review identified mediating variables between financial behavior and its associated consequences. Two mediators that emerged from the literature are financial satisfaction and short-term financial goals.

6) Consequences —In the realm of financial behavior, the consequences encompass financial well-being, depressive symptoms, hindrances in achieving financial goals, foresight in future saving and spending, happiness, investment decisions, long-term financial goals, market discipline, and workplace productivity. Meanwhile, the consequences linked to gamification-related behavioral intention include behavior, learning outcomes, sustained usage, broader system-level changes, perceived impact on learning, use behavior, and user behavioral change. From these findings, it is interesting to examine if gamification may influence financial behavioral intention and manifest into actual financial behavior.

This review highlights potential avenues for future research on financial behavior and gamification-related behavioral intention. Firstly, in the domain of financial behavior, there is a need to delve deeper into socio-economic and demographic antecedents, as suggested by previous studies [15], [36]. Furthermore, the exploration of psychological factors can contribute to a better understanding of financial behavior [29], [44], particularly in relation to the five personality traits: extraversion, agreeableness, conscientiousness, neuroticism, and openness [36], [50]. Additionally, it is important to objectively measure subjective factors such as financial wellness [84], [85] and financial satisfaction [1], [30], [86], [87]. Lastly, investigating the consequences of financial behavior warrants further attention [1], [30].

Regarding gamification-related behavioral intention, future research should aim to develop a comprehensive model that incorporates new design characteristics, user demographics, and contextual factors to gain a deeper understanding of user motivation, experience, and behavior in relation to gamification [57]. Rather than solely focusing on the effectiveness of gamification for users who have already adopted it, researchers are encouraged to explore determinants of gamification success, as suggested by [21]. Furthermore, investigating the potential negative consequences of gamification and methods for mitigation should also be pursued [21].

Future gamification research can also explore the influence of cultural factors [11], [64], [69]–[71] and financial background [72] on gamification adoption. Additionally, investigating constructs such as trust, information asymmetry, perceived risk, and personal innovation holds promise for enhancing understanding of gamification effects [72]. Lastly, specific gamification elements warrant further investigation. For example, [61] suggests future research to explore a new perspective on gamification feedback by examining normative, formative, corrective, positive, and negative feedback [88]–[90].

Methodology

This review identified a range of methods employed in the reviewed studies. The studies focusing on financial behavior primarily utilized quantitative methods, with a predominant use of primary rather than secondary data. Questionnaires were commonly employed to collect primary data, and the analysis of obtained data involved techniques such as structural equation modeling, partial least squares structural equation modeling, and confirmatory and exploratory factor analysis.

For future research on financial behavior, it is recommended to incorporate experimental designs to explore the relationship between personal finances and financial behavior [29]. Additionally, adopting a longitudinal approach would provide valuable insights into how financial behavior and its consequences evolve over time [49]. The utilization of longitudinal and experimental approaches facilitates the identification of direct and indirect relationships between variables [29] and helps mitigate common method bias [43], [44]. Lastly, qualitative approaches are also valuable in enhancing understanding of financial behavior [43].

Table 8. Methodologies of Financial Behavior Research

Source	Data Collection	# (%)	Data Analysis	# (%)	Ref.
Primary data	Questionnaire	26 (81)	Structural equation modeling	7 (22)	[35], [37], [40], [41], [44], [46], [50]
			Partial least squares structural equation modeling	6 (19)	[4], [29], [34], [36], [48], [49]
			Confirmatory factor analysis (CFA)	3 (9.4)	[38], [39], [46]
			Exploratory factor analysis (EFA)	2 (6.3)	[39], [46]
			Covariance-based structural equation modeling (CB-SEM)	1 (3.1)	[15]
			Exploratory mediational analysis with latent change scores	1 (3.1)	[39]
			Generalized structured component analysis (GSCA)	1 (3.1)	[30]
			Logistic regression	1 (3.1)	[41]
			Multi-group analysis (MGA)	1 (3.1)	[35]
			Partial least squares path modeling	1 (3.1)	[42]
			Welch t-test	1 (3.1)	[38]

			Welch analysis of variance	1 (3.1)	[38]
Secondary data	2010 National Longitudinal Survey of Youth 1997 (NLSY97)	1 (3.1)	Structural equation modeling (SEM)	2 (6.3)	[31], [33]
	the 2016 National Financial Well-Being Survey	1 (3.1)			
	the Financial Literacy and Behavior Among Malaysian Young Adults project	1 (3.1)	Covariance-based structural equation modeling	1 (3.1)	[43]
	National Survey on Financial Inclusion 2018	1 (3.1)	Generalized structural equation models (GSEM) with logistic regression basis	1 (3.1)	[32]
	the 2018 FINRA Investor Education Foundation's National Financial Capability Study (NFCS)	1 (3.1)	Stepwise ordinary least squares regression analysis	1 (3.1)	[47]
	the 2017 Survey of Financial Competences of Spanish individuals	1 (3.1)	Univariate and multivariate analysis	1 (3.1)	[45]

Similarly to research on financial behavior, most studies investigating gamification-related behavioral intention also employ quantitative methods. Data collection predominantly relies on questionnaires, while data analysis methods commonly include partial least squares structural equation modeling, structural equation modeling, and confirmatory factor analysis. In addition, a few studies utilize qualitative methods, such as interviews for data collection and axial coding and open coding for analysis. Apart from the empirical studies reviewed, three conceptual studies contribute by presenting research models that elucidate gamification-related behavioral intention.

For future research on gamification-related behavioral intention, it is recommended to involve larger sample sizes and diverse participants to enhance the significance of research findings [53], [54], [57], [60]. Moreover, future studies should gather primary and secondary data from various sources to comprehensively understand the topic [54].

Table 9. Methodologies of Gamification-Related Behavioral Intention

Method	Data Collection	# (%)	Ref	Data Analysis	# (%)	Ref
Quantitative	Questionnaire	25 (71)	[7], [11], [51]–[56], [59]–[64],	Partial least square structural	11 (24)	[7], [54], [76], [56], [61], [66],

			[66]–[76]	equation modeling (PLS-SEM)		[68], [70]–[72], [74]
				Structural equation modeling (SEM)	11 (24)	[11], [51]–[53], [55], [59], [62]–[64], [69], [75]
				Confirmatory factor analysis (CFA)	6 (13)	[11], [52], [55], [64], [69], [73]
				Covariance based approach structural equation modeling (CB-SEM)	1 (2.2)	[73]
				Cross-temporal correlation	1 (2.2)	[53]
				Exploratory factor analysis (EFA)	1 (2.2)	[11]
				Harman’s one-factor analysis	1 (2.2)	[55]
				Hierarchical multiple regression	1 (2.2)	[62]
				Mixed analysis of variance (ANOVA)	1 (2.2)	[62]
				Time-lagged regression	1 (2.2)	[53]
				Variance-based SEM	1 (2.2)	[60]
	Experiment	3 (8.6)	[59], [60], [73]	Analysis of variance (ANOVA)	1 (2.2)	[59]
				Confirmatory factor analysis (CFA)	1 (2.2)	[73]
				Multivariate analysis of variance (MANOVA)	1 (2.2)	[62]

Qualitative	Focus group	1 (2.8)	[66]	Axial coding and open coding	2 (4.4)	[66], [71]
	Interview	2 (5.7)	[57], [71]			
	Diary	1 (2.8)	[57]	Diagrammatic sense-making	1 (2.2)	[57]
Conceptual		3 (8.6)	[58], [65], [77]	Conceptual	3	[58], [65], [77]

D. Conclusion

This systematic literature review focused on the domains of financial behavior and gamification-related behavioral intention, revealing a growing interest in these research areas. This review analyzed 53 articles published from 2018 to 2022 and synthesized them using the TCCM framework to understand the current research landscape comprehensively. In addition to providing insights into theories, contexts, characteristics, and methodologies employed in current research, this review also identifies several areas for future research.

Theoretical Contributions

Gamification research has predominantly emerged from the education and learning domain, while research on gamification in the financial sector remains limited. This review highlights the theories utilized in past research on financial behavior and gamification-related behavioral intention. Given the current lack of a robust theoretical foundation in gamification research, researchers must leverage this understanding to guide future studies. Additionally, there is an opportunity to develop new theories that explore novel aspects of financial behavior and gamification-related behavioral intention. Furthermore, future research should encompass a wider range of contexts, including different countries, cultures, demographics, specific financial apps, and gamification elements.

This review identifies various constructs, including antecedents, mediators, moderators, and consequences, that contribute to understanding financial behavior and gamification-related behavioral intention. This review underscores the need for future research to investigate their relationships by providing a comprehensive overview of these constructs. Furthermore, this review suggests additional constructs to explore in future studies to enhance understanding of financial behavior and gamification-related behavioral intention.

This review also examines the methodologies employed in previous research and provides guidance for future studies to deepen understanding of financial behavior and gamification-related behavioral intention. These methodologies encompass experimental, longitudinal, and qualitative approaches utilizing primary and secondary data from diverse and large samples.

Managerial Implications

This review emphasizes the importance of implementing gamification strategies to enhance behavioral intention. It encourages financial app managers and developers, particularly those involved in PFM apps, to consider integrating gamification elements. Implementing gamification in PFM apps can foster behavioral intention toward various behaviors. The integrated conceptual models presented in this review, outlining key variables and their relationships with financial behavior and gamification-related behavioral intention, offer valuable guidance for app managers and developers in incorporating gamification.

In conclusion, this review provides a comprehensive overview of current financial behavior and gamification-related behavioral intention research. Researchers are encouraged to leverage these findings and pursue future research opportunities to enhance understanding of these domains.

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